Vittorio Maria Ruffo

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Education

Frankfurt School of Finance & Management *Frankfurt, Germany*

PhD in Finance and Master in Business Research and Analytics *September 2022 - present*

- Main research interests: empirical asset pricing, applications of AI, volatility, decentralized finance
- Advisor: Prof. Grigory Vilkov

Collegio Carlo Alberto and University of Turin *Turin, Italy*

Second Level Specializing Master in Finance, Insurance and Risk Management *September 2021 - July 2022*

- Merit based full tuition waiver

University of Verona *Verona, Italy*

Master in Banking and Finance - Quantitative Finance track *September 2019 - September 2021*

- Final Grade: 110/110 with honours
- Supervisor: Prof. Alessandro Bucciol
- Thesis in Econometrics *"Personality and investment choices: an analysis on Australian panel data"*

University of Verona *Verona, Italy*

Bachelor in Economics and Business *September 2016 - September 2019*

- Final Grade: 110/110 with honours
- Supervisor: Prof. Elena Giaretta
- Thesis in Business Management *"Tourism between sustainability and innovation: the "Glamping" phenomenon"*

Working Papers

Market Efficiency in Prediction Markets: A Comparison with Derivatives, with M. Fabi, R. Marfè, and L. Schoenleber [link to paper]

Abstract: We study pricing efficiency in decentralized prediction markets by comparing market-implied probabilities from Polymarket with benchmarks derived from option-implied risk-neutral distributions extracted from the derivatives market. From Bitcoin and Ethereum prediction-market contracts, we find that, although Polymarket prices broadly track option-implied benchmarks, they show systematic price differences driven by behavioral factors and market frictions. Price differences are most pronounced in tail events, during periods of high volatility, and in response to major macroeconomic shocks, and they reflect the influence of sentiment, attention, and blockchain-specific risks. These results reveal both efficiency and behavioral distortions in prediction markets.

Factor Dispersions, with D. Gerchik, L. Schoenleber, and G. Vilkov [link to paper]

Abstract: Dispersion strategies capture the difference in variance dynamics between a basket and its components. Even though smart-beta indices intend to load heavily on a particular factor, factor dispersions based on such baskets are exposed to risks of other factors and idiosyncratic variances. Analyzing factor dispersions through a linear factor model and equicorrelation representations, we recover driving forces behind dispersion dynamics and work out an attribution of a dispersion risk premium. As a balanced combination of systematic and idiosyncratic variance components, dispersion and its risk premium provide signals about future changes in systematic and alpha-based investment opportunities.

- Winner of the CBOE - The Options Institute - S&P Dow Jones Indices Dispersion Research Grant

Work in Progress

Firm-level News Networks, with C. Dim and G. Vilkov

Presentations (* scheduled)

June 2026. (*) **8th Future of Financial Information Conference**, Frankfurt, Germany.
May 2026. (*) **3rd Structured Retail Products and Derivatives Conference**, Hagen, Germany.
May 2026. (*) **Designing DeFi Conference**, New York, USA.
May 2026. **2026 conference Tech 4 Finance 3: AI and Blockchain**, Paris, France.
April 2026. **XXVII Workshop on Quantitative Finance, University di Bergamo**, Bergamo, Italy.
January 2026. **Decentralized Finance & Crypto Workshop, Scuola Normale Superiore**, Pisa, Italy.
December 2025. **2025 Global AI Finance Research Conference**, Hong Kong, China.
July 2025. **Second Liverpool Workshop in Option Markets**, Liverpool, UK.
November 2024. **2024 FMA Conference on Derivatives and Volatility**, Chicago, USA.
May 2024. **Bonn-Frankfurt-Mannheim PhD Conference 2024**, Bonn, Germany.

Teaching and Work Experience

2026-	**Lecturer, Corporate Finance (Bachelor)**, Frankfurt School
2025	**TA in Portfolio Optimization in Continuous Time (Master, Prof. Francesco Sangiorgi)**, Frankfurt School
2025-	**Lecturer, Introduction to Economics and Finance (High school)**, Associazione Lagrange
2025-	**Lecturer, Python Bootcamp (Master), Teaching rating (4.1/5)**, Frankfurt School
2024-	**TA in Financial Products & Modelling (Master, Prof. Grigory Vilkov)**, Frankfurt School
2024-	**TA in Corporate Finance (Bachelor, Prof. Larissa Schaefer and Prof. Yigitcan Karabulut)**, Frankfurt School
2024	**RA for Quantitative Asset Management, Prof. Francesco Sangiorgi**, Frankfurt School
2023	**TA in Portfolio Management (Master, Prof. Paula Cocoma)**, Frankfurt School
2022	**Teacher, Data Analysis Laboratory with R**, University of Verona
2021	**TA in Financial Econometrics (Master, Prof. Alessandro Bucciol)**, University of Verona
2021	**TA in Statistics (Bachelor, Prof. Marco Minozzo)**, University of Verona
2020-2021	**TA in Business Management (Bachelor, Prof. Elena Giaretta and Prof. Angelo Bonfanti)**, University of Verona

Scholarships

2022	**Doctoral Scholarship** Frankfurt School of Finance & Management
2021	**Merit based full tuition waiver** Collegio Carlo Alberto

Computer Skills

Python, Matlab, R, Stata, SQL, Java, Latex, Adobe Lightroom and Photoshop, Phase One Capture One

Additional Information

Languages: English (Fluent), Italian (Native speaker)

Interests: Photography, Hiking, Cooking